Hack-Nocturne '26

Through the night Ideas take flight

Your Idea

SentinelQA

An autonomous AI system for **outcome-driven quality engineering** that uses **Agentic AI, LLMs and Playwright MCP** to plan, execute and monitor software health independently. It bridges the gap between testing and **SRE** by focusing on **automated remediation** and real-time observability rather than manual debugging.

Theme:

- **Agentic AI**
- **Playwright MCP**
- **Full-Stack Observability**

Why should we select you?

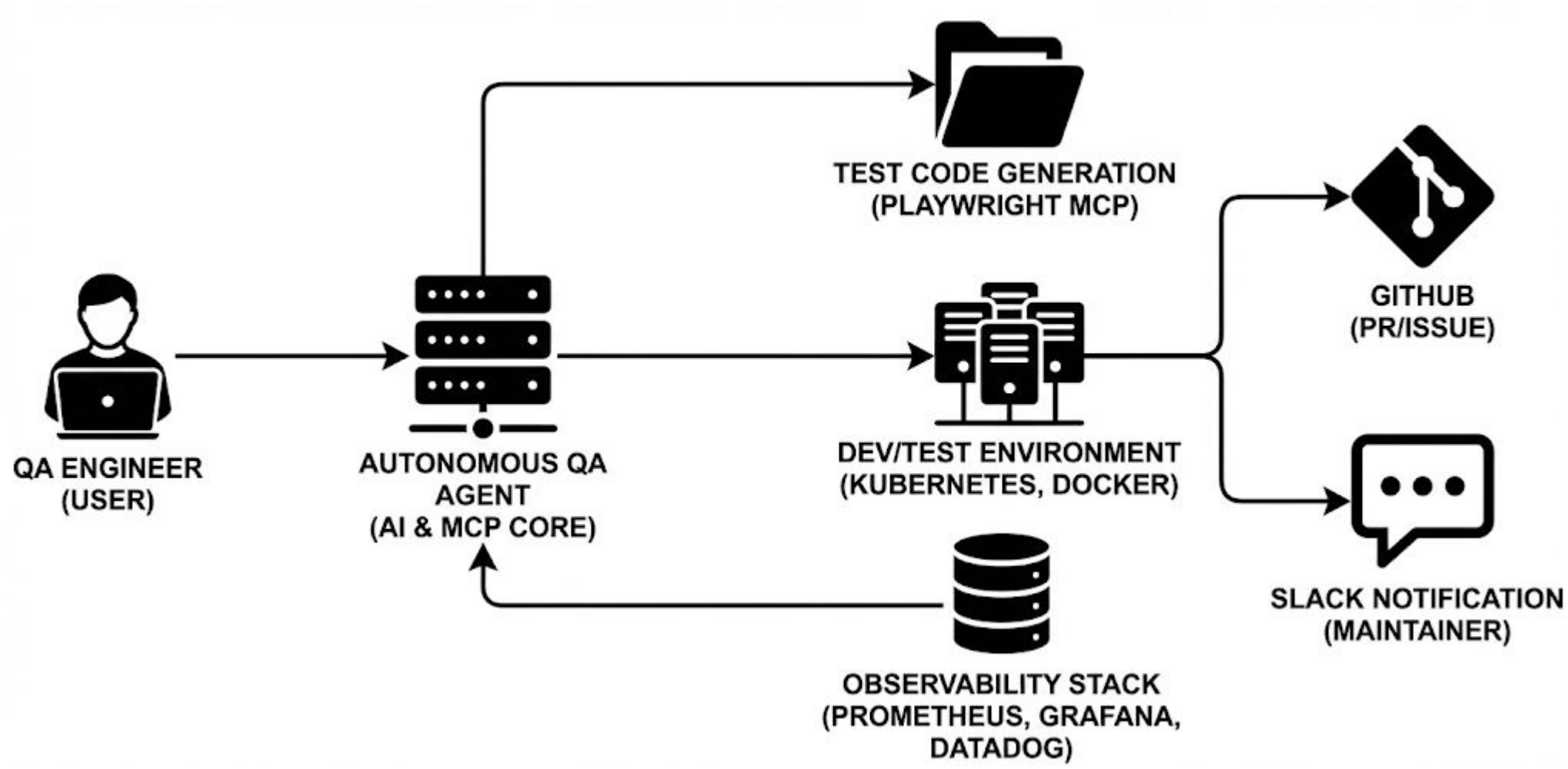
Problem We Are Solving

- **Disconnected Workflows:** Engineering teams are trapped in a **Context Gap** where UI test failures are isolated from infrastructure logs and system metrics.
- **Manual Bottlenecks:** Developers spend **30%** of their time manually triaging errors and correlating failures across **Prometheus**, **Datadog** and **Grafana**.
- **Slow Recovery (MTTR):** Without automated root cause analysis, **push and pray** deployments lead to prolonged downtime and high operational costs.

Our Idea

- **Autonomous Engineering:** An AI system that uses **Playwright MCP** to independently write, execute and heal E2E tests.
- **Observability-Driven QA:** The agent monitors development environments in real-time, querying metrics to perform instant Root Cause Analysis (RCA).
- **Closed-Loop Remediation:** Automatically generates **GitHub Pull Requests** and notifies maintainers via **Slack** with the exact logs needed for a fix.

Architecture Diagram



Tech-Stack & Platforms You Mostly Build On

Frontend / Backend

- **Go:** High-performance backend for agent orchestration and concurrent processing.
- **Node.js:** For seamless integration with the **Model Context Protocol (MCP)** ecosystem.
- **Next.js:** Building a high-end dashboard for real-time test monitoring and PR approvals.
- **MongoDB:** Storing historical test traces, system logs, and agent memory for Root Cause Analysis.

AI & ML Stack

- **Large Language Models (LLMs):** Utilizing **Claude 3.5/4** for complex reasoning and code generation.
- **Agentic AI:** Multi-agent system featuring **Planner**, **Scripter (Playwright)**, and **Healer** agents.
- **Model Context Protocol (MCP):** The core framework for connecting AI agents to external tools like GitHub and Datadog.
- **Vector Databases:** Using **Chroma** or **Pinecone** for semantic search over historical logs and documentation.

Cloud / DevOps / Tools

- **Kubernetes (K8s):** Container orchestration for running headless browsers and monitoring dev environments.
- **Prometheus & Grafana:** Real-time metrics collection to trigger automated remediation loops.
- **Datadog:** Log aggregation and tracing for deep-dive Root Cause Analysis (RCA).
- **Playwright:** The primary engine for autonomous browser testing and DOM inspection.
- **GitHub & Slack API:** Seamless automation of Pull Requests, Issues, and team notifications.

Blockchain & Web3 / Cybersecurity

Leave fields **blank** where **not applicable**, you can put **logos** of tech stacks in the provided space

Thank you

Good Luck G

Team **Hack-Nocturne**
contact@hack-nocturne.in

https://hack-nocturne.in/